Delphes selection vs. the CROWN ntuple baseline

the stage the NSBI test's CMS ntuples correspond to

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Abstract

Comparison of the `--crown` selection applied to the Delphes samples against the ntuple-level selection in `KIT-CMS/BBTauTauAnalysis-CROWN` (`nmssm_config.py`). This is a *different stage* from the published HIG-25-008 analysis selection: CROWN keeps $p_T > 20$ on all three legs and τ_h candidates at the **VVVLoose** working point, applying the trigger thresholds, the elliptical signal region and the categorisation only later. Comparing our analysis-level selection against their ntuple-level one is what produced an apparent factor ∼5 discrepancy in signal efficiency; the correct comparison is this one.

1 Object selection

	CROWN ntuple	**source**	**Delphes** `--crown`	
Muon p_T, $\lvert\eta\rvert$	`tight_muon_min_pt` 20, `max_abs_eta` 2.4	config	20, 2.4	✓
Electron p_T, $\lvert\eta\rvert$	`tight_electron_min_pt` 20, `max_abs_eta` 2.5	config	20, 2.5	✓
τ_h p_T, $\lvert\eta\rvert$	`tight_tau_min_pt` 20, `max_abs_eta` 2.5	config	20, 2.5	✓
τ_h vs-jet WP	**VVVLoose** (`wp: 1`) — *"looser taus needed for tau misidentification estimate"*	config	one **Medium**-equivalent bit	×
τ_h vs-e / vs-μ WP	VVVLoose / VLoose (`wp: 1`)	config	— (single bit)	×
τ_h decay modes	**0, 1, 10, 11** (excludes DM 2)	config	none — Delphes has no decay modes	×
τ_h $\lvert d_z\rvert$	0.2 cm	config	not available	×
Lepton $\lvert d_{xy}\rvert$, $\lvert d_z\rvert$	0.045 / 0.2 cm	paper Tab. 1	not available	×
Lepton ID / isolation	Tight ID + isolation	paper Tab. 1	card efficiency only	×
b-jet p_T, $\lvert\eta\rvert$	`bjet_min_pt` 20, `bjet_max_abs_eta` 2.5 (Run 3)	config	20, 2.5	✓
b-tag requirement	none at ntuple level	inferred†	none (`btag_min 0`)	✓

† No b-tag requirement was found in the configuration; `bjet_min_pt` defines the collection rather than imposing a cut.

2 Pair and event selection

3 Where the residual efficiency gap lives

τ_h working point — the dominant term. CROWN keeps τ_h candidates at VVVLoose and applies Medium later in the analysis, precisely so the misidentification background can be estimated from the looser sample. A Delphes τ_h carries a *single bit* already fixed at a Medium-equivalent working point: there is no looser candidate to keep, because it was never written. DeepTau Medium is ∼70% efficient where VVVLoose is ∼98%, so a factor ∼1.4 on the τ_h leg is expected and cannot be recovered by any selection change. This is the same root cause as the τ_h-object rebuild on the tuning-v2 critical path.

	CROWN ntuple	source	Delphes --crown	
Channel assignment	$\mu \to$ mt, else $e \to$ et, else 2 $\tau_h \to$ tt	paper §5	same priority	✓
Opposite charge	**required**	—	**not applied**: no `Jet.charge` in these ntuples	×
$\Delta R(\ell, \tau_h)$	not found in the configuration	—	> 0.5 applied	?
Extra-lepton veto	`loose_{muon,electron}_min_pt` 10	config	veto counts leptons above **20**	?
Elliptical signal region	**not applied**	—	not applied	✓
Trigger	not at ntuple level	inferred	none	✓

Opposite charge — works the other way. We keep same-sign pairs that CROWN rejects. Same-sign is CMS's QCD control region at ∼99% purity, i.e. an essentially pure fake-τ population, so *our background is inflated* by its absence. Delphes does write `Jet.Charge`; it was simply not carried into the ntuple schema. Now added, and it arrives with the next ntuplisation — no re-simulation required.

Decay modes and impact parameters. Not representable: Delphes has no HPS decay modes and no track impact parameters. Both are acceptance-reducing on the CROWN side, so their absence also inflates our yield slightly.

4 Two rows worth confirming

Both are marked ? above and both are one-line questions:

1. Does CROWN apply a $\Delta R(\ell, \tau_h)$ requirement at ntuple level? None was found in the configuration read; we apply > 0.5, which would make us *tighter* if CROWN does not.
2. What threshold does the extra-lepton veto use? CROWN's loose collections begin at $p_T > 10$, whereas our veto only counts leptons above 20 — so **we keep events CROWN vetoes**, making us looser on this row.

Note that the two point in opposite directions, so they do not simply cancel: one would tighten us and the other loosen us relative to CROWN.

Implementation: `scripts/delphes_to_sbi.py` — `CROWN_SEL` and `cms_select(thresholds=CROWN_SEL)`, reached with `--crown` on both `delphes_to_sbi.py` and `cutflow.py`. Thresholds pinned against the CROWN configuration by tests in `tests/test_cutflow.py`. A companion note, `selection_comparison.tex`, compares the same converter against the published HIG-25-008 *analysis* selection.